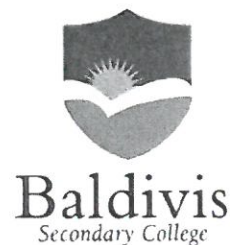


Year 9 - 2015

Investigation 1

Semester One Term 1

Probability



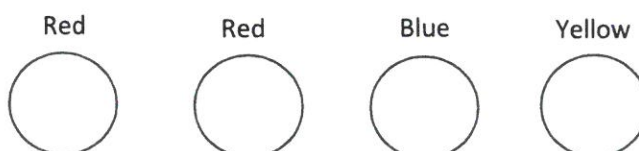
This investigation has three sections. Section 1 will be completed in class as a group. Section 2 is to be completed at home and Section 3 will be completed in class under test conditions. For Sections 1 and 2 you are allowed to ask for help and guidance, however, Section 3 will be completed under test conditions to fully test your understanding. No guidance or hints will be provided.

AIM: In this assessment you will be investigating probabilities of two step events, calculating relative frequencies both with and without replacement. This will be done using both diagrams and calculations.

Section 3

Name: Marking Key. Class: _____ Mark: /18

You have a bag which contains 4 coloured balls: two red, one blue and one yellow.



You choose a ball at random from the bag without replacement. You then choose a second ball and record your outcome.

1. If you were to try this experiment, how many times would you need to complete it to make it valid? Why? (2 marks)

✓ More than 50.

✓ To approach theoretical probability.

2. Draw a tree diagram to show the theoretical probability for each possible choice combination.

(5 marks)



3. Using your tree diagram, find the following probabilities:

(5 marks)

a. Pr (red and blue in any order) $= \frac{4}{12} = \frac{1}{3}$ ✓

b. Pr (yellow then red) $= \frac{2}{12} = \frac{1}{6}$ ✓

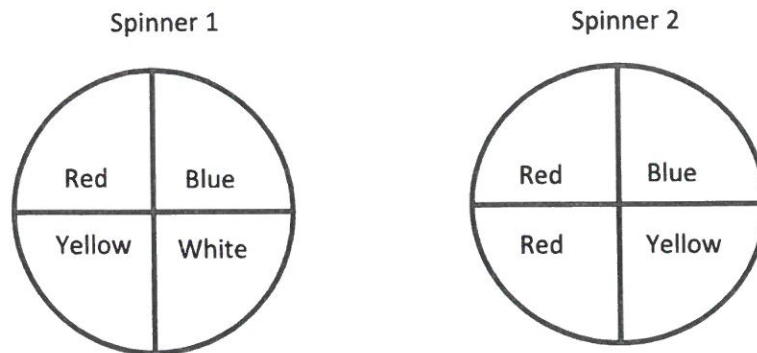
c. Pr (not yellow) $= \frac{6}{12} = \frac{1}{2}$ ✓

d. Pr (not red) $= \frac{2}{12} = \frac{1}{6}$ ✓

e. Pr (yellow or blue on first choice) $= \frac{2}{4} = \frac{1}{2}$ ✓

4. Consider the following frequency table. Is there a difference between the experimental probability (relative frequency) and theoretical probability for this sample space? Why?

(6 marks)



Spinner 1	Spinner 2	Frequency	Relative Frequency
Red	Red	2	$\frac{2}{30} = \frac{1}{15}$
Red	Blue	5	$\frac{5}{30} = \frac{1}{6}$
Red	Yellow	1	$\frac{1}{30}$
Blue	Red	3	$\frac{3}{30} = \frac{1}{10}$
Blue	Blue	2	$\frac{2}{30} = \frac{1}{15}$
Blue	Yellow	4	$\frac{4}{30} = \frac{2}{15}$
Yellow	Red	0	0
Yellow	Blue	1	$\frac{1}{30}$
Yellow	Yellow	3	$\frac{3}{30} = \frac{1}{10}$
White	Red	2	$\frac{2}{30} = \frac{1}{15}$
White	Blue	4	$\frac{4}{30} = \frac{2}{15}$
White	Yellow	3	$\frac{3}{30} = \frac{1}{10}$

Hint: Use this space to calculate the theoretical probability for this sample space.

(3 marks for table)

PTO.

✓ ① for tree diagram.

✓ Yes.

(3 marks total for reasoning)

✓ Experiment not completed enough times to be valid
/ Some outcomes did not occur.

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